Project Title

Evaluating Machine Learning Models for Physiologic Risk Stratification Using Emergency Department Vital Signs at Mercer General Hospital

Statement of the Problem

Triage in emergency departments is critical for prioritising patients based on severity, but many low-resource settings face limitations in data availability and decision-support tools. At Mercer General Hospital, triage nurses currently use the Emergency Severity Index (ESI) system to classify patients; however, there is no structured dataset available that contains clinician-assigned triage labels for research or machine learning development.

This creates a key limitation: while vital signs are routinely collected in the emergency department, there is no existing framework that uses this data alone to systematically identify patterns of patient physiological risk in the local Caribbean population. As a result, most machine learning triage models developed in other countries, such as Araouchi and Adda (2024), cannot be directly applied or validated in this setting due to differences in population characteristics, clinical workflows, and data structure.

This study proposes a 12-week pilot to investigate whether machine learning models can identify meaningful patterns of physiological instability using only routinely collected vital signs. Because no triage labels are available in the dataset, a rule-based physiologic severity classification system will first be constructed using clinically relevant vital sign thresholds. Machine learning models will then be trained to predict these derived severity categories, and performance will be evaluated as a proof-of-concept for data-driven risk stratification in the Caribbean emergency department context.

Previous Work

A series of seven articles discussing the use of Artificial Intelligence (AI) and Machine Learning (ML) in emergency department triage were reviewed, along with one study examining the use of the Canadian Triage and Acuity Scale (CTAS) in a Caribbean hospital. These articles were analysed to gain a better understanding of current triage practices, the potential role of AI and ML in improving patient prioritisation, and the challenges faced within Caribbean healthcare settings.

Araouchi and Adda (2024) developed an AI-based triage system called TriageIntelli to help reduce emergency department overcrowding by improving how patient severity is classified. They used 1,267 patient records from two Korean emergency departments collected between 2016 and 2017 and tested six machine learning models: Support Vector Machines (SVM), Random Forests, Artificial Neural Networks, Gradient Boosting Machines (GBM), Linear Regression, and XGBoost. The Korean Triage and Acuity Scale (KTAS) was used as the classification standard, and SMOTE was applied to balance the dataset. Among the individual models, GBM achieved the highest accuracy at 79%, followed closely by SVM at 78.7%. However, the best results came from a stacking ensemble that combined SVM, GBM, and Linear Regression, reaching an accuracy of 80.05%. This was higher than similar models reported in previous studies. The findings suggest that combining multiple models can improve triage prediction compared to relying on a single model. The authors noted that their results may not be generalisable because they used only one dataset, and they also acknowledged that SMOTE-generated synthetic data may not fully reflect real-world patient cases.

Da'Costa et al. (2025) explored the advantages, challenges, and future applications of AI and machine learning in emergency department triage. Their study was a narrative review of research published between 2014 and 2024, using databases such as PubMed, Scopus, IEEE Xplore, and Google Scholar. The review found that AI can improve patient prioritisation in overcrowded emergency departments, support real-time risk assessment, analyse unstructured information through natural language processing, reduce inconsistencies in triage decisions, and improve resource management during busy periods or mass casualty events. Despite these benefits, the authors highlighted several limitations. Because the study was a narrative review rather than a systematic review or meta-analysis, the overall strength of the evidence could not be measured quantitatively. They also noted that restricting the review to English-language, peer-reviewed articles may have excluded relevant studies, publication bias may have favoured positive findings, and differences in healthcare systems across regions may affect how applicable the results are in different settings.

Tyler et al. (2024) investigated how artificial intelligence and machine learning are being used in emergency department triage. Using the Joanna Briggs Institute scoping review methodology, the researchers searched several databases and selected 29 studies from an initial 1,142 records. The review found that machine learning models generally performed better than traditional triage systems when identifying patient risk, predicting outcomes, and determining whether hospital admission was necessary. AI also showed potential for improving disease detection, resource allocation, and predicting patient length of stay. However, the authors recognised several limitations. Many of the included studies used retrospective data, which may not fully represent real clinical environments. Excluding patients with missing information could also affect the accuracy of AI models, and the study selection process may have introduced bias by potentially overlooking important evidence.

Pooransingh et al. (2018) examined how effectively a modified Canadian Triage and Acuity Scale (CTAS) was being used in the emergency department of Port of Spain General Hospital in Trinidad and Tobago. The researchers collected data from 200 patients over a three-month period and analysed waiting times and demographic factors. They found no significant relationship between triage category and patient demographics, suggesting that patients were treated fairly regardless of age, gender, or ethnicity. However, waiting time targets were not consistently achieved. While low-acuity patients were seen quickly, many high-acuity patients experienced delays, with some waiting much longer than recommended for treatment and hospital bed allocation. The authors linked these delays to staff shortages, limited senior medical support during overnight shifts, and insufficient familiarity with the triage system. They also noted the lack of formal performance monitoring processes. A key limitation was the relatively small sample size, and because the study relied mainly on patient records, it could not assess the clinical reasoning behind triage decisions.

Alomari et al. (2025) investigated whether ChatGPT could accurately and safely perform emergency department triage. In their prospective observational study conducted in Saudi Arabia, 138 patients were assessed by emergency department physicians using CTAS, and ChatGPT was then asked to assign triage scores based on de-identified patient information. Although ChatGPT agreed with physician assessments in 85.61% of cases and showed substantial reliability, its performance dropped significantly when compared with a senior consultant who acted as the gold standard. ChatGPT achieved only 42.86% accuracy against the consultant's decisions and frequently overestimated patient acuity levels. While this tendency may be safer than under-triaging patients, it could place additional pressure on emergency department resources. The authors concluded that ChatGPT may be useful as a decision-support tool but should not be used independently for triage decisions. Limitations included the small sample size, the use of a general-purpose AI model not specifically trained for medical triage, and the inability of the observational design to control for all influencing factors.

Almulihi et al. (2024) conducted a systematic review to evaluate the effectiveness of AI and machine learning in predicting patient outcomes during emergency department triage. After searching several major databases, the authors selected 17 studies from an initial pool of 1,011 articles. The review found that AI and machine learning consistently outperformed traditional triage approaches in areas such as triage accuracy, diagnosis, and early patient management. Most studies reported improved prediction of patient outcomes, and boosting algorithms appeared to deliver the strongest overall performance among the machine learning techniques evaluated. The authors concluded that AI and machine learning have significant potential to improve emergency department care and provide more reliable decision support than traditional scoring systems. However, they also highlighted limitations, including the heavy reliance on observational studies, the combination of studies examining different clinical tasks, and the inclusion of both paediatric and adult populations, which may have affected the consistency of the findings.

Feretzakis et al. (2024) addressed the clinical and operational problem of predicting, at the point of triage, whether an ED patient will require hospital admission or can be safely discharged, with the goal of supporting faster and better-informed disposition decisions in overcrowded emergency settings. Using approximately 280,000 visits from the MIMIC-IV-ED database at Beth Israel Deaconess Medical Centre (2011–2019), the study applied H2O.ai's AutoML platform across 15 candidate models using an 80/20 stratified train-test split, with triage-point variables including vital signs, acuity, age, sex, race, arrival mode, pain score, medications, and waiting time. A Gradient Boosting Machine was selected as the best-performing model, achieving an AUC-ROC of 0.8256, accuracy of 73.17%, and AUCPR of 0.8722, with acuity, waiting hours, current medications, age, and mode of arrival emerging as the most influential predictors. The authors cautioned, however, that the model's moderate recall of 53.08% means a meaningful proportion of patients requiring admission could be missed, and that the absence of external validation limits generalisability beyond the source dataset and population. These constraints are particularly relevant to this study's context: a Caribbean ED serving a demographically and clinically distinct population, where neither the patient characteristics nor the institutional data infrastructure resembles those of a large US academic medical centre.

Gap Analysis

Taken together, the reviewed literature demonstrates that AI and machine learning hold significant promise for improving emergency department triage across diverse settings. However, a consistent limitation across these studies is the narrow geographic scope of validation: the datasets used by Araouchi and Adda (2024), Almulihi et al. (2024), Tyler et al. (2024), Da'Costa et al. (2025), and Feretzakis et al. (2024) are drawn from hospitals in East Asia, the Middle East, Europe, North America, and large academic medical centres, with no representation of Caribbean populations or health systems. Feretzakis et al. (2024) explicitly acknowledge this, noting that their GBM model thta has been trained on a single US institution's data, has not been validated across other hospitals or healthcare systems, and that generalisability remains a key unresolved concern. This is not an isolated caveat; it is a pattern across the literature. As Pooransingh et al. (2018) demonstrated in Trinidad and Tobago, Caribbean emergency departments face distinctive structural challenges — including staff shortages, limited overnight consultant coverage, and inconsistent triage monitoring — that differ substantially from the institutional contexts in which existing AI triage models have been developed and tested. A machine learning model trained on Korean or US academic centre data cannot be assumed to generalise to a small-island Caribbean ED serving a different population with a different disease burden and resource constraints.

A second gap arises from how AI triage performance has been benchmarked in the existing literature. Alomari et al. (2025) is the only reviewed study to compare AI triage decisions against an expert consultant as a gold standard, revealing that ChatGPT's apparent agreement with general physician decisions (85.61%) masked a much lower accuracy when measured against senior clinical expertise (42.86%). Feretzakis et al. (2024) similarly flag that his model's modest recall of 53.08% raises concern about missed admissions, yet this was never tested against consultant-level judgment — only against historical disposition records. This raises a clinically important question about what the appropriate standard of comparison should be for AI triage tools. Yet this type of expert-benchmarked evaluation has not been conducted in a Caribbean emergency department context, where the absence of immediate consultant supervision during overnight shifts means that the gap between routine and expert triage decision-making may be particularly pronounced.

Existing AI triage models have not been validated on Caribbean patient data, and AI triage performance has not been evaluated against expert clinical standards in Caribbean settings. This study begins to address both by training machine learning models on local emergency department data and establishing a baseline performance measure that future work can compare against prospective clinical outcomes and consultant-level assessments. In doing so, it lays the groundwork for a more contextually appropriate and clinically meaningful evaluation of AI-assisted triage at Mercer General Hospital.

Proposed Solution

This study will develop and evaluate machine learning models for physiologic acuity classification using routinely collected vital signs from patients presenting to Mercer General Hospital's emergency department. Because the dataset does not contain clinician-assigned triage levels, a physiologic acuity proxy will first be constructed using a rule-based approach inspired by Emergency Severity Index physiological criteria, particularly those associated with high-acuity presentations (ESI levels 1 and 2). This proxy will represent patient physiological instability rather than a full triage classification system.

The dataset will include standard vital signs recorded at presentation, including the Glasgow Coma Scale (GCS), systolic blood pressure, mean arterial pressure, heart rate, respiratory rate, temperature, and Fraction of Inspired Oxygen (FiO_2). These variables will be used to capture objective indicators of patient severity.

Five machine learning models will be trained on this dataset, namely logistic regression, support vector machines, random forest, gradient boosting machines, and a stacking ensemble model combining the best-performing approaches. The goal is to assess how well machine learning models can learn patterns of physiological instability from routine emergency department data.

Model performance will be evaluated using accuracy, precision, recall, and F1-score. Standard techniques will be used to address class imbalance where necessary. Results will be interpreted in terms of predictive performance and the ability of the models to correctly identify high-acuity patients.

Model performance will be compared to ranges reported in existing literature, including Araouchi and Adda (2024) and Feretzakis et al. (2024), while recognising that direct comparison is limited due to differences in datasets, populations, and label construction methods.

A key limitation of this study is that the target labels are derived from a simplified physiologic rule-based system rather than direct clinician-assigned triage decisions. Therefore, the model is not intended to replicate full Emergency Severity Index decision-making, but rather to explore whether machine learning can identify meaningful patterns of patient severity using vital signs alone.

This study is a 12-week pilot and will not be used for clinical deployment or decision-making. If results are promising, future work will expand the dataset to include additional clinical variables such as presenting complaints, resource utilisation, and clinician-assigned triage categories, enabling full ESI-level prediction and clinical validation in a real-world emergency department setting.

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